数组计数法

人员

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上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3192

王向东老师周六三点半C++数组元素移动

刷新

#	用户名	姓名	编程分	时间	A	B	C	D	E	F	G	H	I	J	K	L
1	chenyiqi	陈奕圻	1200	213	100	100	100	100	100	100	100	100	100	100	100	100
2	lihonghao	李弘浩	1200	232	100	100	100	100	100	100	100	100	100	100	100	100
3	zhaotingshu	赵庭澍	1200	234	100	100	100	100	100	100	100	100	100	100	100	100
4	wangchunshuo	王纯硕	1200	241	100	100	100	100	100	100	100	100	100	100	100	100
5	quyuxuan	屈钰轩	1140	246	100	100	100	100	100	100	100	100	100	40	100	100
6	xiejuntian	解钧添	960	221	100	100	100	100	100	100	100	100	100	60		
7	zhangjiarui	张嘉芮	900	189	100	100	100	100	100	100	100	100	100	0		
8	cuixiaoze	崔孝泽	900	234	100	100	100	100	100	100	100	100	100	0		
9	pengjunwen	彭俊文	800	165	100	100	100	100	100	100		100	100			
10	liuchenxi123	刘晨锡	500	40	100	100	100	100	100	0	0					
11	xuchenxiao123	许辰逍	500	44	100	100	100	100	100							

作业

https://cppoj.kids123code.com/contest/3298 (课上讲了 A ~ F 题, 作业是 F 题必做, G H 选做)

课堂表现

今天上课主要给同学们讲了数组计数法, 关键需要同学们深刻理解 cnt 数组的含义与使用, 建议同学们课后好好复习复习课上的几道题目。

课堂内容

换位置

```
#include <bits/stdc++.h>

using namespace std;

int a[105];

int main()
{
    int n;
    cin >> n;
```

```
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    int maxx = -10000000, minn = 100000000;
    for (int i = 1; i <= n; i++) {
        if (a[i] < minn) {
            minn = a[i];
        }
        if (a[i] > maxx) {
            maxx = a[i];
        }
    }

    int x, y;
    for (int i = 1; i <= n; i++) {
        if (a[i] == minn) {
            x = i;
        }
        if (a[i] == maxx) {
            y = i;
        }
    }

    swap(a[x], a[y]);
    for (int i = 1; i <= n; i++) {
        cout << a[i] << " ";
    }
    return 0;
}
```

移动数组元素

```
#include <bits/stdc++.h>

using namespace std;

int a[105];

int main()
{
    int n;
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    int maxx = -10000000, minn = 100000000;
    for (int i = 1; i <= n; i++) {
        if (a[i] < minn) {
            minn = a[i];
        }
        if (a[i] > maxx) {
```

```
            maxx = a[i];
        }
    }

    int x, y;
    for (int i = 1; i <= n; i++) {
        if (a[i] == minn) {
            x = i;
        }
        if (a[i] == maxx) {
            y = i;
        }
    }

    swap(a[x], a[1]);
    swap(a[y], a[n]);
    for (int i = 1; i <= n; i++) {
        cout << a[i] << " ";
    }
    return 0;
}
```

求n个数中每个数出现的次数

```
#include <bits/stdc++.h>

using namespace std;

int a[105], cnt[15];

int main()
{
    int n;
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    for (int i = 1; i <= n; i++) {
        cnt[a[i]]++;
    }
    for (int i = 1; i <= 10; i++) {
        if (cnt[i] != 0) {
            cout << i << " " << cnt[i] << endl;
        }
    }
    return 0;
}
```

求n个数中出现次数最多的数

```
#include <bits/stdc++.h>

using namespace std;

int a[105], cnt[15];

int main()
{
    int n;
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    for (int i = 1; i <= n; i++) {
        cnt[a[i]]++;
    }

    int maxx = -100000;
    for (int i = 1; i <= 10; i++) {
        if (cnt[i] > maxx) {
            maxx = cnt[i];
        }
    }
    int x;
    for (int i = 1; i <= 10; i++) {
        if (cnt[i] == maxx) {
            x = i;
        }
    }

    cout << x << endl;
    return 0;
}
```

声音识别

```
#include<bits/stdc++.h>
using namespace std;
long long a[100009];
int cnt[105];
int main(){
    long long c,d,f,e,g=0,n;
    cin>>n;
    for(int i=1;i<=n;i++){
        cin>>a[i];
        cnt[a[i]]++;
    }
    for(int i=1;i<=100;i++){
        if(cnt[i]!=0){
            g++;
```

```
        }
    }
    cout<<g<<endl;
    for(int i=1;i<=100;i++){
        if(cnt[i]!=0){
            cout<<i<<" "<<cnt[i]<<endl;
        }
    }
    return 0;
}
```

[NOIP 2005 普及组] 校门外的树

```
#include <bits/stdc++.h>

using namespace std;

int w[10000+5];

int main()
{
  int n, m; cin >> n >> m;
  for (int i = 0; i <= n; i++) {
    w[i] = 1;
  }
  for (int i = 1; i <= m; i++) {
    int l, r; cin >> l >> r;
    for (int j = l; j <= r; j++) {
      w[j] = 0;
    }
  }
  int sum = 0;
  for (int i = 0; i <= n; i++) {
    if (w[i] == 1) {
      sum ++;
    }
  }
  cout << sum << endl;
  return 0;
}
```